Rae Tiffen
Summary of my thesis
May 14, 2026

In my thesis, I apply recursive Frisch-Waugh-Lovell decompositions to find an alternative derivation for omitted variable bias and a total derivative analogue to the regression partial derivative.

Goldberger (1991, p. 186), after introducing the omitted variable bias formula, notes:

> The situation here is quite reminiscent of the distinction between partial and total derivatives in calculus. Indeed $\beta_1 = \beta_{1\cdot 2} + F\beta_{2\cdot 1}$ has the same pattern as $\frac{dy}{dx_1} = \frac{\partial y}{\partial x_1} + (\frac{dx_2}{dx_1})(\frac{\partial y}{\partial x_2})$.

With recursive Frisch-Waugh-Lovell decompositions, I show that Goldberger's comment holds exactly: omitted variable bias is the distinction between the partial and total derivative.

Preliminaries

Partial and total derivatives: Consider a function $f(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$ where $\mathbf{x} = [x_1\ x_2\ \cdots\ x_n]'$. Define the partial and total derivatives of $f(\mathbf{x})$ with respect to x_1:

$$\begin{aligned}\frac{\partial f(\mathbf{x})}{\partial x_1} &:= \lim_{\Delta x_1 \to 0} \frac{f(x_1 + \Delta x_1, x_2, \ldots, x_n) - f(x_1, x_2, \ldots, x_n)}{\Delta x_1} \\ \frac{df(\mathbf{x})}{dx_1} &:= \frac{\partial f(\mathbf{x})}{\partial x_1} + \sum_{i=2}^{n} \frac{\partial f(\mathbf{x})}{\partial x_i} \frac{dx_i}{dx_1}\end{aligned}$$

Note that the definition of the total derivative is recursive.

Frisch-Waugh-Lovell (FWL) decompositions: Consider the OLS-estimated regression of $\mathbf{y} \in \mathbb{R}^n$ on covariate set $\mathbf{X} \in \mathbb{R}^{n \times k}$ yielding projection coefficient $\beta := (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ and $\mathbf{e} := \mathbf{y} - \mathbf{X}\beta$.[1] Define $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2\ \cdots\ \mathbf{x}_k]$ and $\beta = [\beta_1\ \beta_2\ \cdots\ \beta_k]'$.

Denote $\mathbf{X}_{-\mathbf{x_1}} = [\mathbf{x}_2\ \mathbf{x}_3\ \cdots\ \mathbf{x}_k]$ and $\tilde{\mathbf{x}}_1 = \mathbf{x}_1 - \mathbf{X}_{-\mathbf{x_1}}(\mathbf{X}'_{-\mathbf{x_1}}\mathbf{X}_{-\mathbf{x_1}})^{-1}\mathbf{X}'_{-\mathbf{x_1}}\mathbf{x_1}$. By the Frisch-Waugh-Lovell theorem, $\beta_1 = (\tilde{\mathbf{x}}_1'\tilde{\mathbf{x}}_1)^{-1}\tilde{\mathbf{x}}_1'\mathbf{y}$. This result is purely algebraic and does not rely on statistical properties of the estimator. The "Frisch-Waugh-Lovell decomposition of $\mathbf{x}_1$" refers to the regression of $\mathbf{x}_1$ on the set of other covariates, $\mathbf{X}_{-\mathbf{x_1}}$. This decomposes $\mathbf{x}_1$ into the component linearly explained by the set of other covariates and the residual component, $\tilde{\mathbf{x}}_1$. The regression of $\mathbf{y}$ on $\tilde{\mathbf{x}}_1$ yields β_1.

[1] Because my thesis concerns purely algebraic results of the OLS estimator, I do not make the distinction between parameters and estimates. All coefficients discussed here are defined as projection coefficients.

Regression partial derivative: The "partial effect" of $\mathbf{x}_1$ on $\mathbf{y}$ is the difference in predicted $\mathbf{y}$ given a difference in $\mathbf{x}_1$, holding other $\mathbf{x}_i$ fixed. This is equivalently stated as $\frac{\partial \mathbf{y}}{\partial \mathbf{x}_1}$, the partial derivative of $\mathbf{X}\beta + \mathbf{e}$ with respect to $\mathbf{x}_1$. With a model linear in inputs, $\frac{\partial \mathbf{y}}{\partial \mathbf{X}} = \beta$.

Omitted variable bias: Consider the "long" OLS-estimated regression of $\mathbf{y} \in \mathbb{R}^n$ on $\mathbf{X} \in \mathbb{R}^{n \times k}$ and $\mathbf{Z} \in \mathbb{R}^{n \times p}$. Let β_i be the coefficient on $\mathbf{x}_i$ (ith column of $\mathbf{X}$) in this regression, and $\tilde{\beta}_i$ be the coefficient on $\mathbf{x}_i$ in the "short" regression of $\mathbf{y}$ on $\mathbf{X}$ alone (i.e., omitting $\mathbf{Z}$). Then, $\tilde{\beta}_i = \beta_i + \sum_{j=1}^{p} \beta_j \alpha_j$, where β_j is the coefficient on $\mathbf{z}_j$ (jth column of $\mathbf{Z}$) in the long regression and α_j the coefficient on $\mathbf{x}_i$ in the regression of $\mathbf{z}_j$ on $\mathbf{X}$.

Projection operator: I use a system of notation introduced in Yule (1907) and used e.g. in Goldberger (1968). In this system, $\delta_{ab \cdot cd}$ refers to the coefficient on $\mathbf{x}_b$ in the regression of $\mathbf{x}_a$ on $\mathbf{x}_b, \mathbf{x}_c$, and $\mathbf{x}_d$. Similarly, $\mathbf{x}_{b \cdot cd}$ refers to the residuals from the regression of $\mathbf{x}_b$ on $\mathbf{x}_c$ and $\mathbf{x}_d$, that is, the component of $\mathbf{x}_b$ which remains after partialling out the component linearly explained by $\mathbf{x}_c$ and $\mathbf{x}_d$. The $\cdot$ can be viewed as a projection operator. With this notation, the Frisch-Waugh-Lovell theorem can be stated concisely as $\delta_{(a \cdot cd)(b \cdot cd)} = \delta_{a(b \cdot cd)} = \delta_{ab \cdot cd}$. The omitted variable bias result for one omitted variable can be stated as $\delta_{ab} = \delta_{ab \cdot c} + \delta_{ac \cdot b}\delta_{cb}$.

Recursive Frisch-Waugh-Lovell decompositions

Recursive Frisch-Waugh-Lovell decompositions in the 3-covariate case can be stated as $\delta_{ab \cdot cd} = \delta_{a(b \cdot cd)} = \delta_{a(b \cdot (c \cdot d)(d \cdot c))}$.

Consider a regression of the form $\mathbf{y} = \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \beta_3 \mathbf{x}_3 + \mathbf{e}$. From the preliminaries, we know β_1 can be obtained from the regression of $\mathbf{y}$ on $\tilde{\mathbf{x}}_1$, the residuals from the FWL decomposition of $\mathbf{x}_1$. Under this notation, $\tilde{\mathbf{x}}_1 = \mathbf{x}_{1 \cdot 23}$, the residuals from the regression $\mathbf{x}_1 = \delta_{12 \cdot 3}\mathbf{x}_2 + \delta_{13 \cdot 2}\mathbf{x}_3 + \mathbf{x}_{1 \cdot 23}$. Similarly, $\delta_{12 \cdot 3}$ can be obtained from the regression of $\mathbf{x}_1$ on $\mathbf{x}_{2 \cdot 3}$, where $\mathbf{x}_{2 \cdot 3}$ are the residuals from the regression $\mathbf{x}_2 = \delta_{23}\mathbf{x}_3 + \mathbf{x}_{2 \cdot 3}$. Right-hand-side variables in FWL decompositions can themselves be decomposed.

The set of decompositions, and decompositions of decompositions, can be depicted as a tree:

$$\mathbf{y} = \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \beta_3 \mathbf{x}_3 + \mathbf{e}$$

$$\mathbf{x}_1 = \delta_{12 \cdot 3}\mathbf{x}_2 + \delta_{13 \cdot 2}\mathbf{x}_3 + \mathbf{x}_{1 \cdot 23} \qquad \mathbf{x}_2 = \delta_{21 \cdot 3}\mathbf{x}_1 + \delta_{23 \cdot 1}\mathbf{x}_3 + \mathbf{x}_{2 \cdot 13} \qquad \mathbf{x}_3 = \delta_{31 \cdot 2}\mathbf{x}_1 + \delta_{32 \cdot 1}\mathbf{x}_2 + \mathbf{x}_{3 \cdot 12}$$

$$\mathbf{x}_2 = \delta_{23}\mathbf{x}_3 + \mathbf{x}_{2 \cdot 3} \quad \mathbf{x}_3 = \delta_{32}\mathbf{x}_2 + \mathbf{x}_{3 \cdot 2} \quad \mathbf{x}_1 = \delta_{13}\mathbf{x}_3 + \mathbf{x}_{1 \cdot 3} \quad \mathbf{x}_3 = \delta_{31}\mathbf{x}_1 + \mathbf{x}_{3 \cdot 1} \quad \mathbf{x}_1 = \delta_{12}\mathbf{x}_2 + \mathbf{x}_{1 \cdot 2} \quad \mathbf{x}_2 = \delta_{21}\mathbf{x}_1 + \mathbf{x}_{2 \cdot 1}$$

Any regression with k right-hand-side variables can be decomposed into k regressions each with $k-1$ right-hand-side variables, then decomposed into $k-1$ regressions of $k-2$ right-hand-side variables, and so on to the base case of $k!$ bivariate regressions. A fully-decomposed tree contains $\sum_{i=1}^{k} \frac{k!}{(k-i)!}$ coefficients. All coefficients in a tree can be estimated with a series of bivariate regressions, which I implement in Python. The Python code also produces decomposition trees for output with the Forest package in LaTeX.

Regression total derivative and omitted variable bias

Nested functions of this kind hint at applications of the chain rule, which I've found allows for the definition of a regression total derivative and an alternative derivation of omitted variable bias.

By recursive Frisch-Waugh-Lovell decompositions, we can view each variable in terms of their decompositions, recursively to the base case of a series of bivariate regressions. Applying the chain rule to the three-variable tree in the previous section, we find the regression analogue of the total derivative:

$$\begin{aligned}
\frac{d\mathbf{y}}{d\mathbf{x}_1} &= \frac{\partial \mathbf{y}}{\partial \mathbf{x}_1} + \frac{\partial \mathbf{y}}{\partial \mathbf{x}_2}\frac{d\mathbf{x}_2}{d\mathbf{x}_1} + \frac{\partial \mathbf{y}}{\partial \mathbf{x}_3}\frac{d\mathbf{x}_3}{d\mathbf{x}_1} \\
&= \frac{\partial \mathbf{y}}{\partial \mathbf{x}_1} + \frac{\partial \mathbf{y}}{\partial \mathbf{x}_2}\left(\frac{\partial \mathbf{x}_2}{\partial \mathbf{x}_1} + \frac{\partial \mathbf{x}_2}{\partial \mathbf{x}_3}\frac{d\mathbf{x}_3}{d\mathbf{x}_1}\right) + \frac{\partial \mathbf{y}}{\partial \mathbf{x}_3}\left(\frac{\partial \mathbf{x}_3}{\partial \mathbf{x}_1} + \frac{\partial \mathbf{x}_3}{\partial \mathbf{x}_2}\frac{d\mathbf{x}_2}{d\mathbf{x}_1}\right) \\
&= \beta_1 + \beta_2(\delta_{21\cdot 3} + \delta_{23\cdot 1}\delta_{31}) + \beta_3(\delta_{31\cdot 2} + \delta_{32\cdot 1}\delta_{21})
\end{aligned}$$

Applying the omitted variable bias formula from the preliminaries, we can simplify this expression into $\tilde{\beta}_1$, the coefficient from the simple regression of $\mathbf{y}$ on $\mathbf{x}_1$:

$$\begin{aligned}
\frac{d\mathbf{y}}{d\mathbf{x}_1} &= \beta_1 + \beta_2(\delta_{21\cdot 3} + \delta_{23\cdot 1}\delta_{31}) + \beta_3(\delta_{31\cdot 2} + \delta_{32\cdot 1}\delta_{21}) && (1) \\
&= \beta_1 + \beta_2\delta_{21} + \beta_3\delta_{31} && (2) \\
&= \tilde{\beta}_1
\end{aligned}$$

Here, line 1 applies the one-omitted-variable bias formula ($\delta_{ab} = \delta_{ab\cdot c} + \delta_{ac\cdot b}\delta_{cb}$) and line 2 applies the two-omitted-variable bias formula ($\delta_{ab} = \delta_{ab\cdot cd} + \delta_{ac\cdot bd}\delta_{cb} + \delta_{ad\cdot cb}\delta_{db}$).

Because decomposition trees are recursive, this naturally extends to the case of any number of variables. By this application of the chain rule to the decomposition tree, we have defined a regression total derivative and an alternative derivation for omitted variable bias. In doing so, we have confirmed Goldberger's comment, finding that omitted variable bias is exactly the difference between the regression partial and total derivatives.

References

Goldberger, Arthur Stanley (1968). *Topics in Regression Analysis*. New York: Macmillan Publishing.

– (1991). *A Course in Econometrics*. Cambridge, Mass: Harvard University Press. 405 pp. ISBN: 978-0-674-17544-0.

Yule, Udny G. (May 14, 1907). “On the Theory of Correlation for Any Number of Variables, Treated by a New System of Notation”. In: *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character* 79.529, pp. 182–193. ISSN: 0950-1207, 2053-9150. DOI: 10.1098/rspa.1907.0028. URL: https://royalsocietypublishing.org/doi/10.1098/rspa.1907.0028.